

Associations Between Neurophysiological and Cognitive Performance in Patients with Psychosis: an Analysis of the N1, P2 and MMN Event Related Potentials

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Background

- Both cognitive performance and Event Related Potential (ERP) components have been shown to be altered in people with psychosis
- There is evidence for sex differences in the N1/P2 ERPs and some aspects of cognitive performance in healthy controls
- We analyzed relationships between specific domains of cognition (MATRICS) and three ERPs (N1, P2 and MMN) in 38 patients with psychosis to determine whether specific neurophysiological ERP phenotypes correlate with cognitive domains, and whether sex was a mediator of this relationship

Demographics

Table 1

Demographic Information

	SZ	SZA	BPI	p-value
Sample size	N = 8	N = 18	N = 12	
Sex (%)				.284
Female	0	44.44	25	
Male	100	55.56	75	
Mean age ± SD (years)	40.25 ± 12.98	34.78 ± 9.14	29.42 ± 9.34	.073
Race (%)				.600
Caucasian	75	83.33	91.67	
Other	25	16.67	8.33	
Mean education ± SD (years)	14 ± 1.98	15.11 ± 2.08	16 ± 2.11	.421

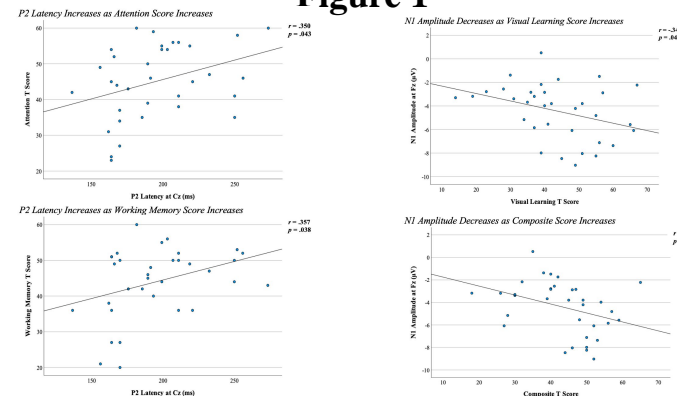
Methods

- Cognition** was assessed with the MATRICS Consensus Cognitive Battery (MCCB) including speed of processing, attention/vigilance, working memory (nonverbal & verbal), verbal learning, visual learning, reasoning and problem solving, and social cognition domains and composite scores
- ERP data** (peaks N1, P2 & MMN) were collected via EEG
 - The N1 and P2 ERPs, elicited using an auditory oddball paradigm, reflect perceptual information registration and processing
 - The MMN, elicited using a duration MMN paradigm, reflects an automatic cortical change-detection process to discriminate changes in a stream of auditory stimulation
- Statistical analyses** included Pearson's correlations, one-way ANOVA and independent samples *t*-tests

Results

- Several cognitive domains were significantly correlated with **N1** and **P2** ERP components (**Figure 1**)
 - P2 latency** was associated with better attention ($r = .350$; $p = .043$), working memory ($r = .357$; $p = .038$), visual learning ($r = .362$; $p = .035$) and verbal learning ($r = .375$; $p = .029$) performance
 - Reduced **N1 amplitude** was associated with worse visual learning performance ($r = -.348$; $p = .044$) and a lower overall cognitive composite score ($r = -.359$; $p = .037$)
- There were no effects of sex or diagnosis on any EEG or cognitive measure

Figure 1



Discussion

- Our results support **N1/P2 being involved in attention-modulated processes** required for cognitive function
- Reduced N1 amplitude may be associated with **decreased visual learning ability**
- Longer P2 latency may index **greater attentional resources required** for stimulus classification processing
- No significant effect of sex or diagnosis was observed on any EEG or cognitive measure

References: Diflorio, A., Jones, I. (2010). Is sex important? Gender differences in bipolar disorder. *International Review of Psychiatry*, 22:5, 437–452.; Grossman, L.S., Harrow, M., Rosen, C., Faull, R., Strauss, G.P. (2008). Sex differences in schizophrenia and other psychotic disorders: a 20-year longitudinal study of psychosis and recovery. *Comprehensive Psychiatry*, 49:6, 523–529.; Schultz, S.H., North, S.W., Shields, C.G. (2007). Schizophrenia: A review. *Am Fam Physician*, 75:12, 1821–1829.

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